Perturbation Catalogue

An Integrated Resource for Exploring the Impact of Genetic Perturbations

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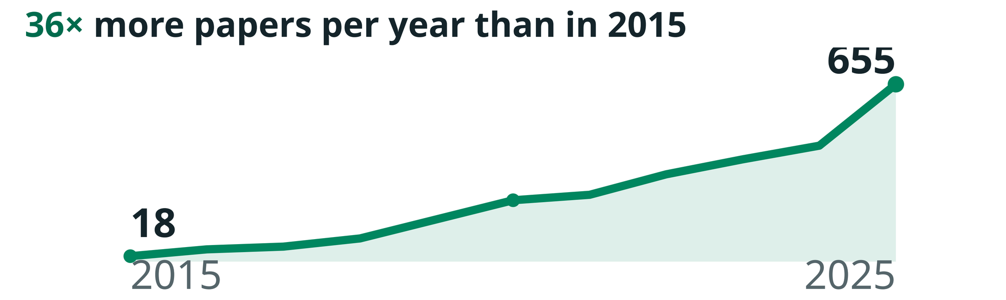
Perturbation evidence is multiplying faster than it can be organised

One genome-scale Perturb-Seq study profiled **more than 2.5 million cells**—but reuse still begins by reconciling incompatible studies.

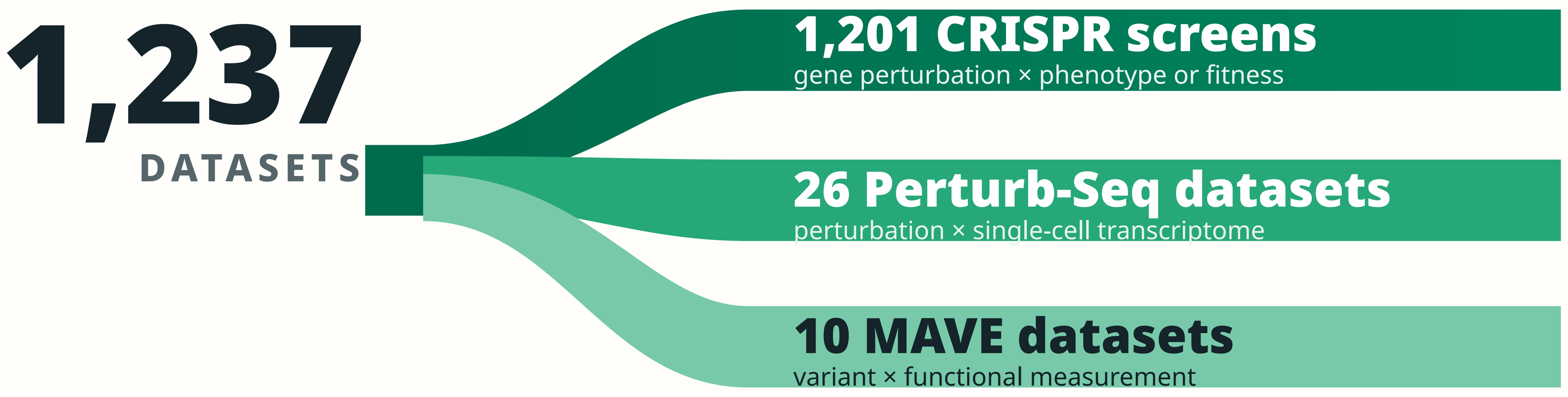
Repositories · identifiers · metadata depth · score semantics · processing choices

36× more papers per year than in 2015



Europe PMC title/abstract query; 24 Aug 2026. Scale example: Replogle et al., Cell 2022.

One Catalogue already connects 1,237 datasets across three complementary modalities



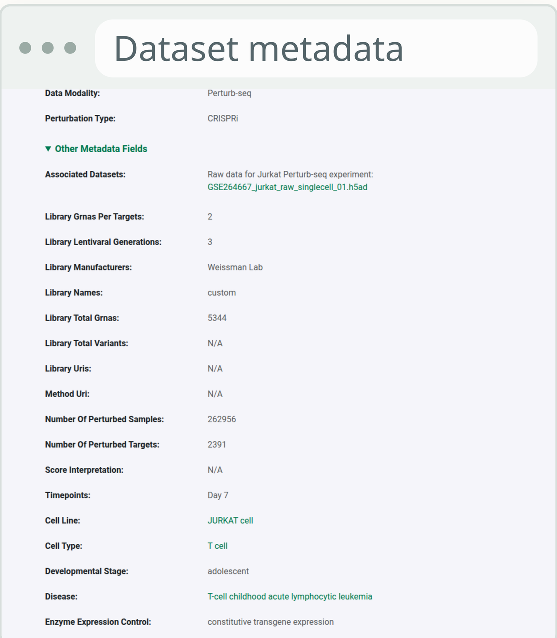
19,235 human targets

36 tissues

34 cell types

1,200 cell lines

Metadata and reanalysis make results comparable



One metadata contract

A validated, extensible schema records the experiment needed to interpret a score—not merely its accession.

Study · perturbation · assay · model system · tissue · cell type or line · disease · timepoint · licence · provenance

Ensembl gene IDs provide stable target identity; biological entities link to established ontologies.

Perturb-Seq reprocessing starts again from raw reads

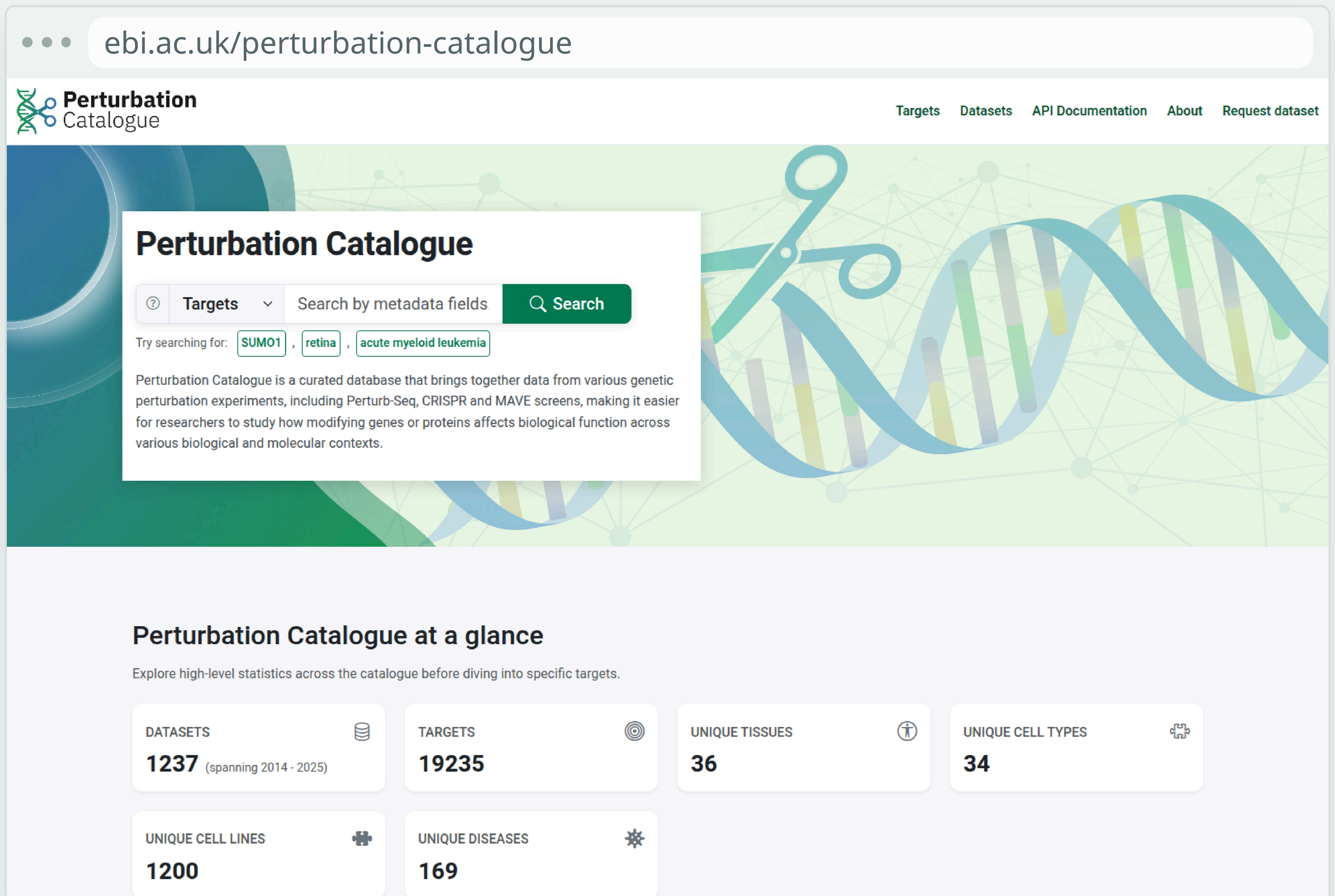
Author matrices differ in alignment, annotation, filtering and guide assignment. Selected studies follow one shared computational path.

SRA / FASTQ — Raw sequencing reads and study-specific probe lists →
Cell × gene counts — Shared reference and gene annotation →
Guide assignment — Map cells to perturbations →
QC + filtering — Produce the final analysis-ready H5AD

Perturbation vs control → **DEA** → **Hallmark GSEA**

Five datasets are live. Broad agreement with author-generated matrices is checked where available; independent quality metrics and wider kit support are next.

Explore the Catalogue in a browser



Target search, current coverage and biological-context summaries in one human-facing portal.

Every result is programmatically accessible

Search, metadata, modality results and complete downloads use the same production backend as the portal.

```
GET /search?query=BRCA2&search_mode=targets
GET /dataset/nadig_2025_jurkat
GET /v1/perturb-seq/nadig_2025_jurkat/search?limit=1
GET /v1/perturb-seq/nadig_2025_jurkat/download?format=csv.gz
```


```
{
  "total_rows_count": 30078802,
  "results": [{
    "perturbation": {
      "gene_symbol": "TFAM",
      "perturbed_target_ensg": "ENSG00000108064"
    },
    "effect": {
      "gene_symbol": "MT-ND5",
      "log2fc": -1.926,
      "padj": 0.0,
      "direction": "decreased"
    }
  }]
}
```


Next, integration becomes continuous and the analytical foundation becomes stronger

Extract structured metadata faster while retaining curator review

AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.

Reanalyse more designs and measure quality independently

More kits and experimental structures, author-independent QC and modality-appropriate methods.

Publish periodic, versioned releases with explicit provenance

Stable snapshots and transparent analytical changes for reproducible downstream use.

Bring us the next hard perturbation-data problem

Suggest datasets, challenge the schema or develop the resource with us. We welcome scientific partnerships and funded multi-year collaborations around your programme's modalities, analyses and functionality.

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Explore the Catalogue

EMBL-EBI

Open Targets